Kshitij Mishra, Ph.D.

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Research Profile

I study how AI agents reason, learn through interaction, and continually improve while remaining consistent, safe, auditable, and aligned with human needs.

- **Reasoning and learning:** post-training, self-improving agents, continual and multi-agent learning, and strategic interaction.
- **Consistency and introspection:** factual consistency, context attribution, and the influence of evidence on agent decisions.
- **Trustworthy AI systems:** safety, security, prompt-injection defense, adversarial evaluation, and auditing of tool-using agents.

CAST research agenda: Consistent, Adaptive, Strategic, and Trustworthy agents.

Appointments and Research Experience

Aug 2025–present
Postdoctoral Associate
Mohamed bin Zayed University of Artificial Intelligence, Abu Dhabi, UAE.
- Research on reasoning, post-training, strategic interaction, and the safety and auditing of foundation-model agents.
- Developing learning methods using textual feedback, process-level rewards, and concept-level reasoning exploration; designing adversarial evaluations and defenses for tool-using agents.

Feb–Jun 2025
Intern
Quantiphi, Bangalore, India.
- Developed two process-level reward formulations to improve reasoning in small language models.

Jan–Oct 2023
Applied Scientist Intern
Amazon.com Services LLC, Los Angeles, USA.
- Worked on language-model editing and privacy-preserving language models, contributing to a Findings of EACL 2024 paper and a granted US patent.

Jul 2019–Aug 2025
Doctoral Researcher
Indian Institute of Technology Patna, India.
- Developed reinforcement-learning and reward-modeling methods for polite, empathetic, persuasive, and personalized conversational systems.

Education

Jul 2019–Aug 2025
Ph.D. in Computer Science and Engineering
Indian Institute of Technology Patna, India. CPI: 8.79.
Thesis: Inducing Politeness and Empathy In User-Oriented Conversational Systems.

Aug 2017–Jun 2019
M.Tech. in Computer Science and Engineering
National Institute of Technology Arunachal Pradesh, India.
CGPA: 9.01; **Gold Medalist**.
Thesis: Heuristic Learning for an Automated Theorem Prover.

Conference Publications

17 conference papers, including accepted work. **Kshitij Mishra** is shown in bold; dotted blue underlining marks first and joint-first authors.

NeurIPS '26
PROBE: Learning to Audit Policy Compliance in Tool-Using LLM Agents
Kshitij Mishra, Abhijith Sharma, Nils Lukas, Salem Lahlou.
NeurIPS 2026. **Accepted**. Code

EMNLP '26 Findings
TRACER: Trace-Regularized Training Against Indirect Prompt Injection
Kshitij Mishra, Salem Lahlou, Nils Lukas.
Findings of EMNLP 2026. **Accepted**. Paper Code

ICML '26
CoRe: Collaborative Reasoning via Cross Teaching
Kshitij Mishra, Mirat Aubakirov, Martin Takac, Nils Lukas, Salem Lahlou.
ICML 2026. **Accepted**. Paper Code

EACL '26 Findings
SD-E^2: Semantic Exploration for Reasoning Under Token Budgets
Kshitij Mishra, Nils Lukas, Salem Lahlou.
Findings of EACL 2026. Paper Code

AAAI '25
Breaking Barriers: A Paradigm Shift in Technology Accessibility for Individuals with Physical Disabilities
Kshitij Mishra, Manisha Burja, Asif Ekbal.
AAAI 2025. Paper Code

COLING '25
MedEx: Enhancing Medical Question-Answering with First-Order Logic based Reasoning and Knowledge Injection
Aizan Zafar, **Kshitij Mishra**, Asif Ekbal.
COLING 2025. Paper Code

EMNLP '24
ABLE: Personalized Disability Support with Politeness and Empathy Integration
Kshitij Mishra, Manisha Burja, Asif Ekbal.
EMNLP 2024. Paper Code

EMNLP '24 Findings
MedLogic-AQA: Enhancing Medical Question Answering with Abstractive Models Focusing on Logical Structures
Aizan Zafar, **Kshitij Mishra**, Asif Ekbal.
Findings of EMNLP 2024. Paper Code

EACL '24 Findings
Correcting Language Model Outputs by Editing Salient Layers
Kshitij Mishra, Tamer Soliman, Anil Ramakrishna, Aram Galstyan, Anoop Kumar.
Findings of EACL 2024. Paper

EMNLP '23
e-THERAPIST: I suggest you to cultivate a mindset of positivity and nurture uplifting thoughts
Kshitij Mishra, Priyanshu Priya, Manisha Burja, Asif Ekbal.
EMNLP 2023. Paper Code

ACL '23
PAL to Lend a Helping Hand: Towards Building an Emotion Adaptive Polite and Empathetic Counseling Conversational Agent
Kshitij Mishra, Priyanshu Priya, Asif Ekbal.
ACL 2023. Paper Code

IJCAI '23
PARTNER: A Persuasive Mental Health and Legal Counselling Dialogue System for Women and Children Crime Victims
Priyanshu Priya, **Kshitij Mishra**, Palak Totala, Asif Ekbal.
IJCAI 2023. Paper Code

Conference Publications (continued)

EACL '23 — **RPTCS: A Reinforced Persona-aware Topic-guiding Conversational System**
Zishan Ahmad, **Kshitij Mishra**, Asif Ekbal, Pushpak Bhattacharyya.
EACL 2023. Paper Code

AAAI '23 — **Help Me Heal: A Reinforced Polite and Empathetic Mental Health and Legal Counseling Dialogue System for Crime Victims**
Kshitij Mishra, Priyanshu Priya, Asif Ekbal.
AAAI 2023. Paper Code

COLING '22 — **PEPDS: A Polite and Empathetic Persuasive Dialogue System for Charity Donation**
Kshitij Mishra, Azlaan Mustafa Samad, Palak Totala, Asif Ekbal.
COLING 2022. Paper Code

NAACL '22 Findings — **Empathetic Persuasion: Reinforcing Empathy and Persuasiveness in Dialogue Systems**
Azlaan Mustafa Samad, **Kshitij Mishra**, Mauajama Firdaus, Asif Ekbal.
Findings of NAACL 2022. Paper

IEEE SMC '19 — **Regularized Universum twin support vector machine for classification of EEG Signal**
Deepak Gupta, Hemanga Jyoti Sarma, **Kshitij Mishra**, Mukesh Prasad.
IEEE SMC 2019. Paper

Journal Publications

6 journal articles. Published research articles, listed by year.

IEEE/ACM TASLP '24 — **Please Donate to Save a Life: Inducing Politeness to Handle Resistance in Persuasive Dialogue Agents**
Kshitij Mishra, Mauajama Firdaus, Asif Ekbal.
IEEE/ACM Transactions on Audio, Speech, and Language Processing (2024), 32, 2202–2212. Paper Code

PLOS ONE '23 — **GenPADS: Reinforcing politeness in an end-to-end dialogue system**
Kshitij Mishra, Mauajama Firdaus, Asif Ekbal.
PLOS ONE (2023), 18(1), e0278323. Paper Code

IEEE TCSS '23 — **Predicting Politeness Variations in Goal-Oriented Conversations**
Kshitij Mishra, Mauajama Firdaus, Asif Ekbal.
IEEE Transactions on Computational Social Systems (2023), 10(3), 1095–1104. Paper Code

Neurocomp. '22 — **Please be polite: Towards building a politeness adaptive dialogue system for goal-oriented conversations**
Kshitij Mishra, Mauajama Firdaus, Asif Ekbal.
Neurocomputing (2022), 494, 242–254. Paper Code

Environ. Earth Sci. '21 — **Artificial intelligence for suspended sediment load prediction: a review**
Deepak Gupta, Barenya Bikash Hazarika, Mohanadhas Berlin, Usha Mary Sharma, **Kshitij Mishra**.
Environmental Earth Sciences (2021), 80, article 346. Paper

Research Cell '19 — **Artificial Neural Networks in Hydrology**
Kshitij Mishra, Deepak Gupta.
Research Cell: An International Journal of Engineering Science (2019), 31, 52–59 (Special Issue, January). Paper

Patent

Systems and methods for updating large language models

Kshitij Mishra, Tamer Soliman, Aram Galstyan, Anoop Kumar, and Anil K Ramakrishna.

US Patent 12,456,020 B1. Granted October 28, 2025. Amazon Technologies, Inc. Patent record

Deployed Research Systems

Etihad Conversational Booking Agent. Designed, developed, and deployed a conversational agent for natural-language flight search and booking assistance at MBZUAI.

Sevak, an Indian-language chatbot. Worked with two fellow researchers to deploy 14 multilingual conversational systems covering railways, judiciary, and healthcare during my Ph.D. at IIT Patna.

Research software: CoRe • SD-E2 • PEPDS • PAL • ABLE • EDiSS

Teaching and Mentoring

Indian Institute of Technology Patna

Supported teaching in theory of computation, machine learning, natural language processing, and artificial intelligence. Mentored students across academic levels and contributed to research on conversational AI.

Academic Service

2022 Organizer, GIAN workshop on *Deep Learning Techniques for Conversational Artificial Intelligence*, IIT Patna, April 13–24.

2020 Organizer, CEP course on *Deep Learning for NLP*, IIT Patna, January 10–23.

2018 Organizer, workshop on *Recent Trends in Computing and Computer Networking*, NIT Arunachal Pradesh, September 3–7.

2017 Organizer, workshop on *Recent Trends in Machine Learning, Big Data and IoT*, NIT Arunachal Pradesh, November 13–17.

Honors and Fellowships

2019 **Prime Minister's Research Fellowship**, Government of India.

2019 **M.Tech. Gold Medal**, NIT Arunachal Pradesh.

2017 **GATE Scholarship**, Government of India.

Technical Skills and Languages

Programming: Python, C++, C, LaTeX.

Frameworks: PyTorch, TensorFlow, scikit-learn, Hugging Face, Google Colab.

Languages: English and Hindi.

Additional Training

2019 *Machine Learning for Engineering and Science Application*, IIT Madras.

2018 *Introduction to Machine Learning*, IIT Kharagpur.